

William Causey

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EDUCATION

Kennesaw State University

Marietta, GA

Bachelor of Science in Software Engineering

Graduation Date: May 2027

- **Related Coursework:** Data Structures and Algorithms, Discrete Structures, Database Systems, Probability and Data Analysis, Computer Organization and Architecture

PROJECTS

RAG Pipeline | AWS Lambda, S3, DynamoDB, Pinecone, OpenAI, Supabase, Vite

April 2026

- Architected and deployed a multi-tenant RAG SaaS application using AWS Lambda, S3, and DynamoDB, implementing automated PDF text extraction, chunking, and OpenAI vector embeddings via Pinecone for semantic document retrieval
- Engineered token-based authentication with Supabase Auth using API-gateway-managed keys with instant revocation and audit logging, enforcing per-user data isolation across S3, DynamoDB, and Pinecone vector metadata to prevent cross-tenant data exposure
- Developed a full-stack responsive frontend using Vite and JavaScript featuring real-time streaming LLM responses, drag-and-drop PDF upload, and document filtering, deployed on Vercel with CI/CD via GitHub
- Implemented serverless security controls including user-scoped rate limiting, IAM least-privilege policies, file validation, and environment-based secret management across AWS and Vercel infrastructure
- Live Demo: rag-pipeline-frontend.vercel.app/

C++ Terminal Dashboard | C++23, CMake, Win32 API, GoogleTest

April 2026

- Built a multi-threaded system monitoring tool in C++ that tracks CPU, memory, and disk usage in real time with <50ms update latency using Win32 APIs
- Implemented asynchronous API integration with libcurl to fetch geolocation and weather data without blocking UI rendering
- Designed thread-safe architecture using mutexes and shared state synchronization.
- Automated testing and deployment via GitHub Actions, enabling automated builds and unit testing
- GitHub Repo: github.com/notwilliam1/terminal-dashboard

Competitive Draft Discord Bot | Python, Discord.py, Asyncio, OOP

October 2025

- Designed and shipped an asynchronous, event-driven matchmaking bot in Python using Discord.py and Asyncio, handling concurrent multi-user interactions with OOP-structured player state, draft logic, and match workflows
- GitHub Repo: github.com/notwilliam1/8s-draft-discord-bot

TECHNICAL SKILLS

Languages: Java, Python, C++, SQL(SQLite, PostgreSQL), JavaScript

Libraries & Frameworks: libcurl, nlohmann/json, FTXUI, Asyncio, Discord.py

Tools: Git, Docker, AWS Lambda, S3, DynamoDB, CMake, GoogleTest, Linux, Pinecone, Node.js

Practices: Agile/Scrum, RESTful APIs, Object-Oriented Design, Memory Management, Multithreading, Concurrency, Unit Testing, Serverless Architecture, CI/CD, Prompt Engineering